

r/Coronavirus and the Viral Pandemic

A source of truth, comfort and companionship during COVID19

J. A. Cole*

Royal Holloway University of London
Jennifer.cole@rhul.ac.uk

C. Van de Ven

University of Amsterdam
casvandeven2000@gmail.com

C. M. Sharkey

Office of the Assistant Secretary for
Preparedness and Response, U.S.
Department of Health and Human
Services
phdsharkey@gmail.com

P. Doherty

University of Illinois at
Urbana-Champaign
pdoherty31@gmail.com

P. Williamson

School of Environmental Sciences,
University of East Anglia
p.williamson@uea.ac.uk

F. Robb

Science Media Centre
freya@sciencemediacentre.org

ABSTRACT

The COVID19 pandemic has unfolded alongside a concurrent ‘infodemic’ – defined by the World Health Organization as an overabundance of information, some accurate, some not, that occurs during an epidemic. Key to managing this is not only identifying, countering and debunking misinformation but also providing unbiased and factually correct information and signposting people towards it. However, during COVID19, the ‘truth’ has not always been clear. It has not always been easy to prepare public health messaging that is consistent, easily understood or practical for everyone to apply. This presents unique challenges, to which social media platforms need to be part of the solution. One such solution can be found on www.reddit.com where, in January 2020, a group of research scientists, students, academics and medics came together to create and moderate forums in which the pandemic can be discussed and questions about it answered. These forums provide case studies of how information can be generated, misinformation corrected and disinformation debunked on subreddits with, combined, more than 3 million subscribers.

CCS CONCEPTS

• Human Computer Interaction; • Collaborative and Social Computing; • Network Architectures;

KEYWORDS

Infodemic, Public health messaging, misinformation

ACM Reference Format:

J. A. Cole, C. Van de Ven, C. M. Sharkey, P. Doherty, P. Williamson, and F. Robb. 2021. r/Coronavirus and the Viral Pandemic: A source of truth, comfort

*Corresponding author: Jennifer Cole, PhD, Department of Geography, Royal Holloway University of London, Egham, Surrey TW20 0EX. Tel: 00(44) (0)7471 996317.

Permission to make digital or hard copies of part or all of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for third-party components of this work must be honored. For all other uses, contact the owner/author(s).

CHI '21 Extended Abstracts, May 08–13, 2021, Yokohama, Japan

© 2021 Copyright held by the owner/author(s).

ACM ISBN 978-1-4503-8095-9/21/05.

<https://doi.org/10.1145/3411763.3450396>

and companionship during COVID19. In *CHI Conference on Human Factors in Computing Systems Extended Abstracts (CHI '21 Extended Abstracts)*, May 08–13, 2021, Yokohama, Japan. ACM, New York, NY, USA, 6 pages. <https://doi.org/10.1145/3411763.3450396>

1 INTRODUCTION

“I cannot thank you and the other mod[erators]s enough for doing this. You have been a source of truth, comfort, and companionship for a lot of us who had no-one else to face the darkness with.”

– u/ArcadianDelSol, r/coronavirus 19 November 2020

As the COVID19 pandemic has spread across the world, a concurrent pandemic of information has spread with it. Deemed an ‘infodemic’ by the World Health Organization and described as, “an overabundance of information, some accurate, some not, that occurs during an epidemic” [1], this proliferation of data, research and opinions has provided both opportunities and challenges. On 15 February 2020, WHO Director-General Tedros Adhanom Ghebreyesus called for the need to manage the COVID19 infodemic, including the need to form strategic partnerships with social media and technology platforms and stakeholders from academia and civil society [2]. A month prior to this, a group of research scientists, academics, students, computer scientists and experienced moderators had come together on the aggregator platform www.reddit.com to manage the infodemic from within. The forums they built, including r/coronavirus (for general news and discussion), r/COVID19 (for discussion of the latest scientific papers) and r/COVID19_support (for emotional support and questions regarding risk and risk management) had grown to more than 2.75 million subscribers by the end of December 2020: more than the official number of deaths from the disease at that time. In mid-2020, r/coronavirus attracted more than 600 million page views per month from 20 million unique users. r/COVID19_support attracted between 500k and 1 million page views per month during the same period; r/COVID19 attracted 10 million page views. The forums curate information provided by anyone who wants to contribute on a platform that is accessible to everyone with an internet enabled device, reasonable bandwidth and a browser, and a reasonable level of English-language literacy.

Governments have tended to approach the infodemic through a focus on the misinformation and disinformation that spreads through informal networks and social media, seeing only the challenges associated with social media platforms and calling for top-down approaches to information management [3]. The flow of information on peer-to-peer platforms and the opportunities such platforms provide for discourse also offers opportunities, however, particularly for embracing knowledge exchange and enabling dissemination through an emergent bottom-up approach that can manage good information, counter misinformation and thus build trust. Doing so depends on understanding the ways in which information is exchanged over such platforms. In this paper, we – representatives of the moderator teams of the most popular reddit coronavirus forums and the scientific and communications specialists whose expertise we have drawn on – argue that the people best placed to help others navigate the infodemic are those who are immersed within it. The best way to manage the infodemic is to bring the people who hold expert knowledge to the places where the largest number of people seek it out and discuss it – i.e., on the world's most popular social media platforms.

Primarily, understanding how people seek, find, process and act on information is dependent on understanding how that information fits with communicative ecologies: the socio-cultural framing and analysis of the local context in which communication occurs and which shapes its meaning [4]. In the digital age, public health messages that will help to prevent and contain COVID19 can be disseminated quickly, as can messages that are able to counter misinformation before it is able to take hold. Information – both good and bad – spreads through populations in patterns similar to those seen in the spread of disease [5], offering opportunities to push out information on protective behaviour ahead of the virus, and so provide at-risk populations with valuable knowledge and tools for fighting it. WHO has identified five key roles in infodemic management: scan and verify evidence; explain the science; amplify the reach of messages; measure the infodemic and assess trends and impacts; and co-ordinate and govern information technology [1]. Social media platforms are key stakeholders in the production, interpretation and dissemination of knowledge. Their role in the COVID19 infodemic is especially important because media – whether traditional or social – is neither ‘good’ nor ‘bad’, high quality nor low. Each medium is unique and shaped by the information it mediates. The better the quality of the information a platform is given, the better the quality of the information it can host and disseminate. Furthermore, individuals evaluate multiple sources; they do not see ‘social media’ as a single source of information; they differentiate between messages passed by friends and family and those from official agencies and expert voices. Reddit attracts these official and expert voices and its network architecture helps to signpost users towards them [6].

1.1 Understanding What Information People Want and When They Seek It

During a public health emergency, people seek out more information, more actively, and from more sources, than they did beforehand [6]; they want to learn more about the disease. The more credible they consider the source, the more likely the information is

to be believed and acted on [7]. How people process such information, differs, however: some prefer to read and self-educate, while others prefer to ask questions and discuss answers in a more socially interactive environment [8]. Some people struggle to process complex concepts; they may be insufficiently practiced at critical thinking, and vulnerable to taking sensational headlines at face value. Information needs to be not only provided but also, sometimes, unpacked and clearly explained. A question-and-answer session with experts is one format in which to provide further explanation of terms and data that may not be easily understood or have been misunderstood. Reddit holds regular AMAs – ‘Ask Me Anything’ – live sessions. The coronavirus forums have held AMAs with Bill Gates, with volunteers on the Pfizer and Moderna vaccine trials, WebMD, PolitiFact, and key scientists working on the response to the pandemic; their engagement has been critical to the forums’ success. But users can also ask the moderators, and one another.

A person’s desire to ask questions increases the closer disease risk comes and the more immediate the individual perceives the risk to be [6]. The granularity of the answers required also increases. Experts contributing to question-and-answer sessions can enable more context-specific questions to be answered – regarding the risks associated with certain types of workplace, for example, or the risks to those with certain underlying health conditions. Even when there is abundant information from national public health authorities and international organizations, individuals may be uncertain about how to apply that guidance to the specific circumstances of their own lives. Information seekers prefer to be able to engage directly with experts, but if experts are not immediately available, they will take whatever information they can find. In August 2020, biosafety professionals partnered with r/COVID19_support to disseminate key messages through the forum and have since provided hundreds of context-specific answers and information regarding workplace, school and leisure facility safety. When such expert advice is provided through peer-to-peer forums, the answers given remain on the forum and so can be searched for and read by others in the same situation.

1.2 Don’t Let the Perfect Be the Enemy of the Good.

During the emergence of a new public health threat like the COVID19 pandemic, no-one is a complete expert. A scientist who has spent a lifetime studying other viruses, or even non-viral pathogens and other health emergencies, is likely to know more about managing COVID19 risks and the likely trajectory of the pandemic than most people, however, as is a historian or economist who has studied past outbreaks. The best guess of such experienced experts even if they are not SARS-CoV-2 experts, may be the best option available. Such people must not be afraid to offer what expertise they have as conspiracy theorists, anti-vaccination supporters, lockdown sceptics and mask opposers will happily fill any information gaps the experts don’t. Misinformation thrives in a vacuum. WHO infodemic training leans heavily on Voltaire’s famous aphorism, already popular within healthcare: “don’t let the perfect be the enemy of the good” [9].

Experts on relevant and closely related subjects need to be prepared to say what they do know, even if it's not everything: for example, they can explain which mutations may change the game, and which are less likely to; why case fatality rate estimates are likely to be much higher at the start of an epidemic than once better data becomes available; and why it is not unusual for vaccine trials to pause while additional safety checks are assessed. Even people with partial expertise can help to explain why uncertainty at stages of a scientific process does not indicate failure.

When a single expert is not available, the next best option is to ask a crowd [10]. This method was widely used by the American military during the Cold War, when the RAND Corporation developed the Delphi Method [11], a means by which an expert group – all of whom knew part of the answer – could collate and consolidate that knowledge to arrive at the best available option. The process was first employed to consider how the world might react to a nuclear war but the approach is well-suited to an open, egalitarian and transparent discussion forum such as reddit. Adding another perspective, management theory often points to the value of making snap decisions based upon the best available information, stressing the importance of speaking quickly whilst acknowledging that if you do, you must also be brave enough to retract if later proven wrong. It is better for the best-informed people to say something: if they don't, there is a danger that less informed, or even wild guesses are sucked into the vacuum and allowed to circulate unchallenged.

1.3 Consider How Research and Guidance Might Be Interpreted By Non-Academic Audiences

During a pandemic, interest in science and scientific publications increases dramatically. The public, newspapers and television channels become much more interested in academic research and findings can be misinterpreted, particularly as the public may ask questions that experts don't anticipate. For example, in April 2020, a non-peer reviewed paper suggested that aerosols from joggers and cyclists had the potential to spread further than the 1.5m social distance recommended by many governments [12]. The lead author had not anticipated that a study on particle aerodynamics would be interpreted as being about infection risk, nor that it would attract considerable media attention, but it did. Similarly, papers on the dangers posed by mutations in the SARS-CoV-2 virus tend to be sensationalized in the media, giving a misleading impression of their likely impact. The science journal *Nature* has countered this with a more accurate description [13] but it is not easily available or accessible to all readers. When engaging with social media, experts need to anticipate how the science might look to a non-scientist, including how it might be misunderstood, and try to head this off with clear and straightforward explanations from the start.

During a pandemic an abundance of new – often tailored – guidance is available from academic, national, and international bodies (e.g. U.S. Centers for Disease Control and Prevention; U.K. National Health Service; World Health Organization), directed at informing local and individual decisions to mitigate the impacts of the disease. This guidance can be difficult for the public to interpret and to apply to individual situations, including: complicated guidelines

for isolation and quarantine, dependent on several variables such as duration of exposure, a positive or negative COVID19 test, and whether the contact is a household member or not; the circumstances that should lead a person to seek testing; or how to reduce risk of transmission within a household. Experts with perspectives on applying this nuanced information can play a large role in helping to assure the effective and responsible implementation of safety guidelines.

1.4 Don't Just Say Something's Wrong – Explain Why It's Wrong: Counter, Correct And Balance.

Spelling out the ways in which research and scientific data has been misinterpreted by the media is a modern-day version of the famous military mantra “know your enemy” [14] – in this case, not the SARS-CoV-2 virus, or the channels and mediums through which misinformation about it might spread but existing narratives, the science (il)literacy of the person who generated it and the receiving reader. The originator might genuinely believe there to be a government cover-up of numbers, or that vaccine development is dangerously rushed, or that masks are ineffective. Rather than ignoring or deleting imperfect information, forum managers should be encouraged to look for whether and where else it has been debunked, as linking to it can be used to respond, deflect others, and arm the community with better information. Websites such as www.fullfact.org and www.snopes.com monitor circulating misinformation and counter it. News agencies such as the UK's Science Media Centre connect responsible media outlets with scientists and researchers who can answer questions, discuss circulating narratives and support and enable more accurate reporting, from which everyone benefits. Social media platform users can take what they produce and arm their users with it, enabling them to spread it wider.

1.5 Social Media Can Fill Information Vacuums With the Best Information Available

During COVID19, there have been many successful examples of experts engaging with the public directly through social media, for example, the epidemiologist Dr Trevor Bedford, who engages on Twitter, and nursing academic Dr John Campbell on YouTube. The moderator team of r/coronavirus, r/COVID19 and r/COVID19_support includes research scientists, medical doctors, nurses, mental health counsellors, academics in fields including virology, epidemiology, paediatrics, public health, resilience, emergency communications, political science and biosafety professionals. One of the moderators, Jennifer Cole, was one of the first cohort of WHO infodemic managers.

Through social media, experts can challenge misconceptions directly, providing platform users with better information to challenge things they see elsewhere. The voting systems on websites such as reddit act as a form of quality control. Previous research has shown such voting mechanisms do improve the quality of information and ensure that the better information rises to the top on platforms where information is peer-curated [6]. A good example from reddit is the response to a preprint posted on the academic archiving site BioRxiv [15] and linked to reddit on 1 Feb 2020, which

implied that SARS-CoV-2 may have characteristics suggestive of artificial origin. Speculation on reddit was quashed not only by comments that immediately replied to the original post but also by the scientists in the moderator team, who quickly wrote and posted a formal response [15]; in the process, they recruited two additional moderators from among the many other scientists who engaged with the discussion to debunk the original paper.

There is value in debunking rather than simply ignoring or removing imperfect information. Here, it is important to understand how such information has arisen. A concerned citizen may think they are genuinely uncovering facts others are trying to hide. They may be open to explanations from scientists who understand how novel viruses emerge, to biosecurity experts who understand how difficult it would be to manufacture a biological weapon, and to political scientists who can see no immediate motive for doing so. The citizen is more likely to listen to explanations that directly address their concerns rather than ridiculing them. The presence of a forum where people can discuss these questions with experts can engender trust that is difficult to establish with one-way communication pushed out from public health agencies. People, and especially younger people, are exposed to a wide range of viewpoints online; some may not undertake critical fact-checking and simply accept what they see at face value, but many are more critical than this. Especially when information sources are suspected to be less reliable, answers are validated between different sources and believed only when there is consensus [6]. This kind of engagement has particular value where misinformation is coming from a source that should be trusted – such as a national leader: former U.S. President Trump was identified as a key source of COVID19 misinformation [17].

2 CONCLUSIONS

We are living in an interconnected world in which the infodemic recognizes no boundaries [2]. Many scientists and academics have already risen to the challenge, making suggestions of how to fight it, producing frameworks for action [2], and undergoing WHO training in infodemic management [18]. Leading academics [19] and entire academic journals [20] took a political stance against misinformation spread by Donald Trump, stressing the greater value of Joe Biden's pro-science approach to the pandemic and giving a clear signal to the U.S. public who the scientists and academics thought was best placed to steer their country through the most serious pandemic of their lifetime. Academics and experts have a similar role to play but must remain mindful of the dangers of politicizing science. Such people work with evidence and facts, shape them into knowledge and send that knowledge out into the world where it changes lives, shapes the future and empowers others. The way knowledge is collected, discussed, shaped, stored, disseminated and accessed is constantly evolving: academics and scientists must evolve with it. We must ride the infodemic wave so that, just as with the COVID19 pandemic itself, we stay ahead of, and flatten, its curve.

[*] This paper is a modified version of Cole, J. The COVID-19 infodemic: the role and place of academics in communicating science to the public. *Global Journal of Medicine and Public Health*, vol. 9, no. 2, 2020.

REFERENCES

- [1] Viroj Tangcharoensathien, Neville Calleja, Tim Nguyen, Tina Purnat, Marcelo D'Agostino, Sebastian Garcia-Saiso S, Mark Landry, Rashidian Arash, Clayton Hamilton, AbdelHalim AbdAllah, Ioan Ghiga, Alexandra Hill, Daniel Hougen-dobler, Judith van Andel, Mark Nunn, Ian Brooks, Pier Luigi Sacco, Manilo de Domenico, Philip Mai, Anatoliy Gruz, Alexandre Alaphilippe and Sylvie Brand. 2020. Framework for managing the COVID-19 infodemic: methods and results of an online, crowdsourced WHO technical consultation. *Journal of Medical Internet Research* (June 2020), 22(6):e19659. DOI: <http://doi.org/10.2196/19569>
- [2] World Health Organization. 2020. Risk communication and community engagement readiness and response to coronavirus disease (COVID-19): interim guidance, 19 March 2020. Retrieved from [https://www.who.int/publications-detail/risk-communication-and-community-engagement-readiness-and-initial-response-for-novel-coronaviruses-\(ncov\)](https://www.who.int/publications-detail/risk-communication-and-community-engagement-readiness-and-initial-response-for-novel-coronaviruses-(ncov))
- [3] Rupert Jayant Limaye, Molly Sauer, Joseph Ali, Justin Berstein, Brian Wahl, Anne Barnhill and Alan Labrique. 2020. Building Trust While Influencing Online COVID-19 Content in the Social Media World. *The Lancet*. (June 2020) 2(6):e277-278 DOI: [https://doi.org/10.1016/S2598-7500\(20\)30084-4](https://doi.org/10.1016/S2598-7500(20)30084-4)
- [4] Greg Hearn and Marcus Foth. 2007. Communicative ecologies: Editorial preface. *Electronic Journal of Communication* (2007) 17(1-2).
- [5] Sebastian Funk and Vincent A. A. Jansen. 2012. The talk of the town: modelling the spread of information and changes in behaviour. In: *Modeling the interplay between human behavior and the spread of infectious diseases*. Springer, New York, NY. (November 2012) 93-102.
- [6] Jennifer Cole. 2017. *The Role of Online Discussion Forums During a Public Health Emergency*. Doctoral Thesis. Royal Holloway, University of London (RHUL), London, United Kingdom
- [7] C. Nadine Wathen and Jacquelyn Burkell. 2001. Believe it or not: Factors influencing credibility on the Web. *Journal of the American society for information science and technology* (December 2001) 53(2) 134-144. DOI: <https://doi.org/10.1002/asi.10016>
- [8] Neil. D. Fleming. 2001. Teaching and learning styles: VARK strategies. Christchurch, New Zealand.
- [9] Klim McPherson. 1994. The Cochrane Lecture. The best and the enemy of the good: randomised controlled trials, uncertainty, and assessing the role of patient choice in medical decision making. *Journal of Epidemiology & Community Health* (1994) 48(1), 6-15.
- [10] Tanja Aitamurto. 2016. Collective Intelligence in Law Reforms: When the Logic of the Crowds and the Logic of Policymaking Collide. In 49th Hawaii International Conference on System Sciences (HICSS), January 5-8, 2016, Kauai, Hawaii, USA. IEEE DOI: <https://doi.org/10.1109/HICSS.2016.349>
- [11] Norman Dalkey. 1969. An experimental study of group opinion: the Delphi method. *Futures* (September 1969) 1(5), 408-426. DOI: [https://doi.org/10.1016/S0016-3287\(69\)80025-X](https://doi.org/10.1016/S0016-3287(69)80025-X)
- [12] Bert Blocken, Fabio Malizia, Thijs van Druenen and Thierry Marchal. 2020. Towards aerodynamically equivalent COVID19 1.5 m social distancing for walking and running. April, 2020, Retrieved February 13, 2021 from <http://www.urbanphysics.net/COVID19>
- [13] Ewen Callaway. 2020. The coronavirus is mutating-does it matter? *Nature*: 585(7824):174-177. DOI: <https://doi.org/10.1038/d41586-020-02554-6>
- [14] Daniel N. Maxwell, Trish M. Perl and James B. Cutrell (2020). "The Art of War" in the Era of Coronavirus Disease 2019 (COVID-19). *Clinical infectious diseases*. (October 2020) 71(16) 2215-2217. DOI: <https://doi.org/10.1093/cid/ciaa229>
- [15] Prashant Pradhan, Ashutosh Kumar Pandey, Akhilesh Mishra, Parul Gupta, Praveen Kumar Tripathi, Manjot Balakrishnan Menon, James Gomes, Perumal Vivekanandan, and Bishwajit Kundu. 2020. Uncanny similarity of unique inserts in the 2019-nCoV spike protein to HIV-1 gp120 and Gag. Retrieved from www.biorxiv.org February 13, 2020. DOI: <https://doi.org/10.1101/20.01.30.92.78.71>
- [16] Discussion of paper retrieved November 18 2020 from www.reddit.com/r/Coronavirus/comments/ewxne6/an_analysis_of_the_recent_paper_about_hiv/
- [17] Sarah Evanega., Mark Lynas, Jordan Adams, Karinne Smolensky, K. 2020. Coronavirus misinformation: quantifying sources and themes in the COVID-19 'infodemic'. Retrieved November 18, 2020 from https://allianceforscience.cornell.edu/wp-content/uploads/2020/10/Evanega-et-al-Coronavirus-misinformation-submitted_07_23_20.pdf
- [18] World Health Organization. 2020. 1st WHO Infodemic Management Training. Retrieved December 11, 2020 from <https://www.who.int/teams/risk-communication/infodemic-management/1st-who-training-in-infodemic-management>
- [19] Howards Frumkin and Sam S. Myers. 2020. Planetary health and the 2020 US election. *The Lancet* (October 2020) 396(10257) 1048-1050. DOI: [https://doi.org/10.1016/S0140-6736\(20\)32038-9](https://doi.org/10.1016/S0140-6736(20)32038-9)
- [20] Scientific American Editors. 2020. Scientific American endorses Joe Biden: We've never backed a presidential candidate in our 175 year history—until now. *Scientific American*. Retrieved December 11 2020 from www.scientificamerican.com/article/scientific-american-endorses-joe-biden/

A A PANEL DISCUSSION WITH THE SESSION AUDIENCE: HOW DID THE REDDIT CORONAVIRUS FORUMS DO, AND COULD WE DO BETTER?

A.1 Set the Scene (15 minutes)

The session will begin with a 15-minute overview of the paper outlined above, followed by a 60-minute Question and Answer session in which the panelists will describe their experiences of engaging with traditional and social media during the COVID19 pandemic, highlighting challenges and opportunities and acknowledging barriers and threats. We will ask questions such as how can professional organizations be encouraged to engage more with social media? Is there a particular role for apolitical professional organizations in building trust on social media platforms?

A.2 Aims and Objectives (15 minutes)

The 60-minute panel discussion that follows the presentation will enable two-way interaction with the audience on the positive and negative aspects of managing and engaging with social media platforms for public health messaging. Audience participation will be enhanced through use of real-time voting and survey tools such as Slido. We will:

- 1) Inform the participants of what we see happening on social media, and the explain inner workings of reddit, highlighting the unique characteristics of reddit as a good social media platform that only works well if it's moderated.
- 2) Ask: is there a danger of leaving the moderation to volunteers? What could reddit (and other platforms) do better?
- 3) Ask: What a good tool for public health information should look like? What works well and less well?
- 4) Ask: Will the current situation, in which experts and scientists are often suspicious of social media, be different in 20-30 years' time when an older generation has grown up with social media and is less suspicious of it?

A.3 Debate with the Audience (15 minutes)

The audience will be encouraged to debate:

- Whose responsibility it is to ensure that users of social media platforms – including Facebook, Twitter and others, as well as reddit – receive reliable, timely and truthful information? Should the platforms do more to control content?
This discussion will use examples of different challenges encountered by the panelists (e.g. verifying 'citizen journalism' YouTube videos purporting to be from inside locked-down Wuhan; debunking the Indian lab origin paper; handling politicized HCQ discussions; deciding how to handle the Great Barrington Declaration). How were these challenges dealt with? Does the audience think the panelists did well and what could be done better?
- How should platforms such as reddit deal with forums such as r/LockdownSkepticism or r/Wuhan_Flu that display an entrenched political position but may be difficult to classify outright as misinformation, rather than highly biased and selective opinion? How does this impact on human rights and freedom of speech?

- How does the COVID19 situation align with Climate Change denial and what can we learn from this?

The panelists will discuss challenges including science denial and climate skepticism. Science denial threatens to gain power on the internet, as do other divisive politics such as white supremacy and anti-Semitism. Populist politics are often meme/soundbite friendly – how are they best countered? How do platform moderators manage information when there is still unclear and incomplete data from outside laboratory settings to, for example, 'prove' masks work?

A.4 Ask the Audience (15 minutes)

- 1) What social media does the audience use (Slido voting)? What did they think of the quality of information on it?
- 2) People need context to be able to navigate information, particularly scientific information. Has the audience had trouble navigating this and if so, what with specifically?
- 3) Discuss how we, the moderators, judge what is right and what is wrong. How do we determine what is 'fake news'? How do we judge which posts stay up and which come down? What are we justified in taking down? What should be removed and what should be allowed under freedom of speech?
- 4) We argue that reddit is good because it allows discussion. Should we therefore avoid removing misinformation at all, and instead allow it to stay and be discussed? What are the advantages and disadvantages of this?

A.5 Looking into the Future (15 minutes)

What lessons have we learned from the COVID19 pandemic on how experts should engage with social media platforms and how does this influence the future?

B PANEL BIOGRAPHIES

B.1 Panel organizer: Jennifer Cole, PhD, Royal Holloway, University of London, UK

Jennifer Cole is a Research Fellow at Royal Holloway University of London, UK. Her PhD thesis explored the role of discussion forums as a source of public health information. From 2007-2017, she worked at the Royal United Services Institute, a UK think tank, where she worked on the cyber security policies and resilience policies of the UK Cabinet Office and NATO, including pandemic flu planning. She is a World Health Organization Infodemic Manager. In January 2020, she volunteered as a moderator of r/coronavirus, the largest pandemic discussion forum on www.reddit.com, which now has 2.4 million subscribers; of r/COVID19, a specialist forum for discussing the latest scientific research; and created r/COVID19_support, which offers advice and emotional support to people struggling with the impact of the pandemic.

B.2 Panelist: Patrick Doherty MSc, University of Illinois at Urbana-Champaign, USA

Patrick Doherty received his MSc from the University of Notre Dame, where he studied tuberculosis and HIV co-infections. He is currently working in the biotech industry. He previously received

his BSc from the University of Illinois at Urbana-Champaign in Molecular and Cellular Biology and is currently working on his MBA. His work, in industry and research, has revolved around the development of vaccines and therapeutics for the treatment of infectious diseases such as *M. tuberculosis*, *Y. pestis*, and various influenza strains including H5N1. He has extensive experience working in BSL3 level labs, as well as aerosol exposures in animal models. His work as a Reddit moderator on r/coronavirus began shortly after other moderators noticed his debunking of a misinformed academic pre-print that had gained traction on the internet by wrongly associating similarities between HIV and SARS-Cov-2.

B.3 Panelist: Freya Robb BA (Hons), Science Media Centre, UK

Freya Robb is a Press Officer at the UK Science Media Centre (SMC), an independent press office that helps to ensure the public have access to the best scientific evidence and expertise through the news media when science hits the headlines. The focus is the UK mainstream media (broadcast, print and online). The SMC works with scientists to help them communicate their research and expertise on the biggest and most contentious stories of the day, whether this is climate change, diet and nutrition, natural disasters, infectious diseases or e-cigarettes. Obviously 2020 has been all about COVID-19. The SMC has run more than 100 press briefings and issued thousands of comments from experts on the big news stories or research of the day – covering basic virology, vaccine and drug development, evidence around non-pharmaceutical interventions, modelling, clinical treatments, policy decisions and much more, keeping journalists – and as a result the public – up to date and well informed of the rapidly developing evidence around COVID-19.

B.4 Panelist: C. Matthew Sharkey, PhD, U.S. Department of Health and Human Services, USA

Matt Sharkey is a Biologist in the Office of the Assistant Secretary for Preparedness and Response at the U.S. Department of Health and Human Services (HHS). He has a PhD in Biochemistry and

Molecular Biology. He was a BSL-3 laboratory virologist for more than nine years and has worked as a reference member of the U.S. Laboratory Response Network, a biosafety officer in a SARS-CoV and MERS-CoV laboratory, and a policy analyst. He is a lecturer in biosafety, biosecurity, and biodefense at Georgetown University. Since August 2020 he has assisted r/COVID19_support, offering context for the sometimes difficult-to-navigate public health guidance to redditors. The views he expresses are his own and are not representative of the views of the U.S. government or any of its agencies or departments.

B.5 Panelist: Cas van de Ven, University of Amsterdam, The Netherlands

Cas van de Ven is a 20-year-old political science undergraduate from the Netherlands. He has been a moderator on the r/coronavirus subreddit for a year and can shed light on how the moderators try to arm the public with accurate, trustworthy and nuanced information. While not a medical expert himself – he is a political science student – he can still provide a platform to academics and the general public where valuable engagement can take place. A prime example is the numerous AMAs he has helped to organize, which enable experts – such as Bill Gates and WebMD – to answer questions from the general forum users. He will discuss why this engagement is important, how the moderators try to deal with misinformation and how other social media can help to manage information spread in this pandemic.

B.6 Panelist: Phil Williamson, PhD, School of Environmental Sciences, University of East Anglia, UK

Phil Williamson has mostly worked for the UK Natural Environment Research Council (NERC, now part of UK Research and Innovation), based at the University of East Anglia with particular interests in science coordination and communication. His main expertise is in global climate change – how it affects natural systems and society, and what we can do about it. His TV interviews on climate change include those for BBC World News, Sky News and CNBC, also radio interviews for BBC R1, R2 and R4, and many regional radio stations.